

Scalability & Future Plan

Date	12 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S. No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Model precision/recall on the reject class is modest ($\approx 42\%$ / $\approx 40\%$) due to severe class imbalance ($\sim 1.7\%$ rejections) in the source data.	Some risky applicants may be missed, and some safe applicants may be flagged; result page discloses this so it is used as one input, not a final verdict.	High
2	Model is trained on a static CSV snapshot and loaded once at startup; there is no automated retraining pipeline.	Predictions can drift as real applicant/credit patterns change over time.	Medium
3	No authentication, request logging, or rate limiting on the Flask app.	Not yet suitable for direct multi-user production exposure without hardening.	High

Scalability Plan:

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single-process local Flask development server (python app.py), suitable for demo/local use.	Deploy behind a production WSGI server (Gunicorn/uWSGI) with a reverse proxy (Nginx), and containerize with Docker for horizontal scaling.
2	Data Storage	Historical applicant and credit-record data stored as flat CSV files (cleaned_credit_data_v3_vintage.csv).	Migrate to a managed relational database (e.g., PostgreSQL / MySQL) or a cloud data warehouse to support larger datasets and incremental updates.
3	Performance	Model loaded once at startup; single synchronous prediction per request.	Add response caching for repeated inputs, and move to an async request pipeline or a dedicated model-serving layer (e.g., IBM Watson Machine Learning, FastAPI + Uvicorn) for higher concurrency.
4	Security	No authentication; applicant data is not persisted after a request completes.	Add HTTPS termination, request authentication for analyst users, input validation hardening, and audit logging of

S. No	Scalability Aspect	Current State	Proposed Upgrade / Solution
			predictions for compliance.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 1	Cloud deployment of the Flask app (e.g., Render, Heroku, or IBM Cloud) with a public URL and HTTPS.	1–2 months	Enables real users/evaluators to access the tool without a local setup.
Phase 2	Automated model retraining pipeline on a refreshed data snapshot, with versioned model artifacts.	2–4 months	Keeps predictions aligned with current applicant and credit-risk patterns.
Phase 3	Add user authentication and role-based access (customer vs. bank analyst/administrator).	3–5 months	Supports differentiated access: customers submit and view predictions; administrators manage the deployed model.
Phase 4	Explainability layer (e.g., SHAP values) surfaced on the result page to show which factors drove a given prediction.	4–6 months	Improves trust and actionability, letting rejected applicants understand what to improve.